

Auto Installer Genius

Automated Remote Software Installation Tool

Installation Manual | Version 1.0

Version	1.0
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Platform	Windows 10/11 (local) to Ubuntu 20.04 / 22.04 / 24.04 (remote)
AI Engine	Cerebras qwen-3-235b-a22b-instruct-2507
Repository	https://github.com/vishnu-18/auto-installer

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1. System Requirements

Component	Minimum Requirement
Operating System	Windows 10 or Windows 11 (64-bit)
Python	3.10 or higher
RAM	4 GB (8 GB recommended)
Disk Space	200 MB free (excluding log files)
Network	Internet access for Cerebras AI API calls
Remote Target	Ubuntu 20.04 / 22.04 / 24.04 LTS
SSH on Remote	OpenSSH server running on port 22 (default)
Remote User	SSH user must have sudo privileges

2. Prerequisites

Before installing Auto Installer Genius, ensure the following are ready:

On your Windows machine:

- Python 3.10 or higher installed and added to PATH.
- pip (Python package manager) available in the terminal.
- Git installed (optional, for cloning from GitHub).
- A Cerebras account with an active API key (free tier is sufficient).

On the remote Ubuntu machine:

- OpenSSH server installed and running.
- The SSH user account must have sudo privileges.
- Network connectivity from your Windows machine to the Ubuntu machine.
- Port 22 open (or your custom SSH port).

Note: No additional software needs to be pre-installed on the remote Ubuntu machine. The application handles everything over SSH.

3. Downloading the Application

Option A - Clone from GitHub (recommended)

Open a Command Prompt or PowerShell and run:

```
git clone https://github.com/vishnu-18/auto-installer.git
cd auto-installer
```

Option B - Download ZIP from GitHub

- Go to <https://github.com/vishnu-18/auto-installer>
- Click the green Code button, then Download ZIP.
- Extract the ZIP to a folder, e.g. F:\MySws\auto-installerV1.0\

The application folder should contain these files:

File	Description
main.py	Application entry point - run this to start the app.
gui.py	GUI layout and event handlers.
installer.py	Core installation orchestration logic.
ssh_client.py	SSH connection and command execution.
ai_client.py	Cerebras AI API integration.
config_manager.py	Save/load .mc.json configuration files.
requirements.txt	Python package dependencies.
.env.example	Template for the .env API key file.

4. Installing Python Dependencies

Open a Command Prompt or PowerShell, navigate to the application folder, and run:

```
pip install -r requirements.txt
```

This installs the following packages:

Package	Version	Purpose
paramiko	>=3.0	SSH connection and SFTP file upload.
python-dotenv	>=1.0	Load API keys from the .env file.
requests	>=2.31	HTTP calls to the Cerebras AI API.
tkinter	built-in	GUI framework (included with Python).
reportlab	>=4.0	PDF generation (for manuals only).

To verify the installation was successful:

```
python -c "import paramiko, dotenv, requests; print('All OK!')"
```

Note: If you see 'ModuleNotFoundError' for tkinter, reinstall Python and make sure the 'tcl/tk and IDLE' option is checked during installation.

5. Getting a Cerebras API Key

Auto Installer Genius uses the Cerebras AI API to generate and verify installation commands. A free account gives access to the models used by this app.

Steps to get your API key:

- **Step 1:** Go to <https://cloud.cerebras.ai> and sign up for a free account.
- **Step 2:** Log in and navigate to the **API Keys** section.
- **Step 3:** Click **Generate New Key**.
- **Step 4:** Copy the key immediately - it will not be shown again.

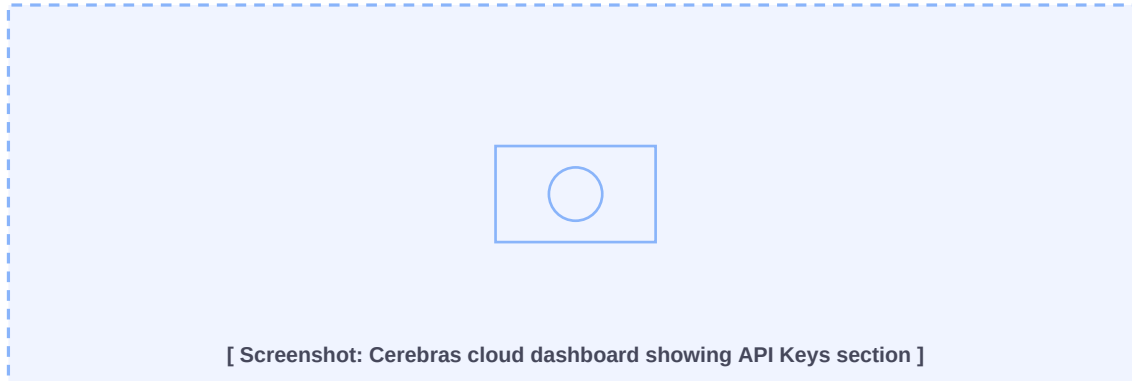


Fig 5.1 - Cerebras API Keys page (cloud.cerebras.ai)

The free tier includes access to:

- llama3.1-8b - smaller, faster model.
- qwen-3-235b-a22b-instruct-2507 - larger model used by this application.

Warning: Keep your API key private. Never commit it to version control or share it. The .gitignore already excludes the .env file.

6. Configuring the API Key

There are two ways to add your Cerebras API key to the application:

Option A - Via the Settings menu (recommended)

- Launch the application: `python main.py`
- Click **Settings** in the menu bar.
- Click **Environment Variables (.env)...**
- The .env editor opens. Add or update the line:

```
CEREBRAS_API_KEY=csk-xxxxxxxxxxxxxxxxxxxxxx
```

- Click **Save**. The key is reloaded immediately.

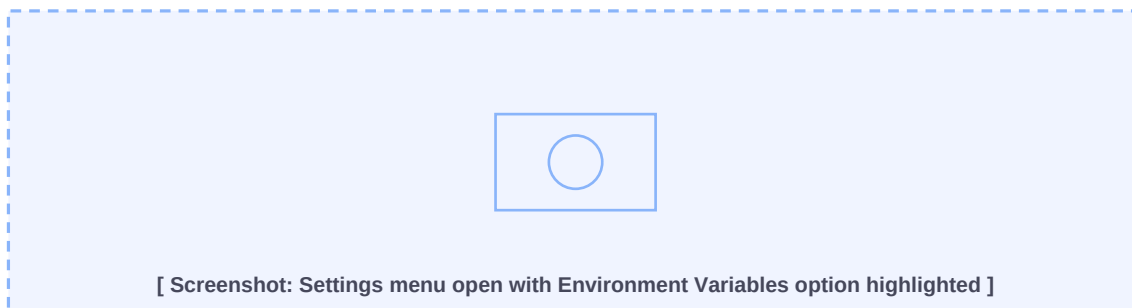


Fig 6.1 - Settings menu and .env editor

Option B - Edit the .env file directly

Copy the .env.example file to .env and edit it:

```
copy .env.example .env
notepad .env
```

Set the following values (no quotes around values):

```
CEREBRAS_API_KEY=csk-xxxxxxxxxxxxxxxxxxxxxx
CEREBRAS_MODEL=qwen-3-235b-a22b-instruct-2507
```

Note: The CEREBRAS_MODEL line is optional. If omitted, the app defaults to qwen-3-235b-a22b-instruct-2507 automatically.

7. Preparing the Remote Ubuntu Machine

The target Ubuntu machine needs only OpenSSH server and a sudo-enabled user. Run these commands directly on the Ubuntu machine (or via any existing SSH session):

7.1 Install and enable SSH server

```
sudo apt-get update
sudo apt-get install -y openssh-server
sudo systemctl enable ssh
sudo systemctl start ssh
```

7.2 Verify SSH is running

```
sudo systemctl status ssh
```

You should see **Active: active (running)** in the output.

7.3 Ensure the user has sudo privileges

```
# Check current groups
groups your_username

# Add to sudo group if not already there
sudo usermod -aG sudo your_username
```

7.4 Find the machine's IP address

```
ip addr show | grep 'inet ' | grep -v 127.0.0.1
```

Note the IP address shown (e.g. 192.168.100.197). This is what you enter in the **Remote IP / Host** field.

7.5 Test SSH from your Windows machine

Open PowerShell and test the connection:

```
ssh your_username@192.168.100.197
```

Note: The application uses SUDO_ASKPASS to handle sudo non-interactively. No changes to /etc/sudoers are required on the remote machine.

8. Running the Application

Navigate to the application folder and run:

```
python main.py
```

The GUI window will open and the application is ready to use.

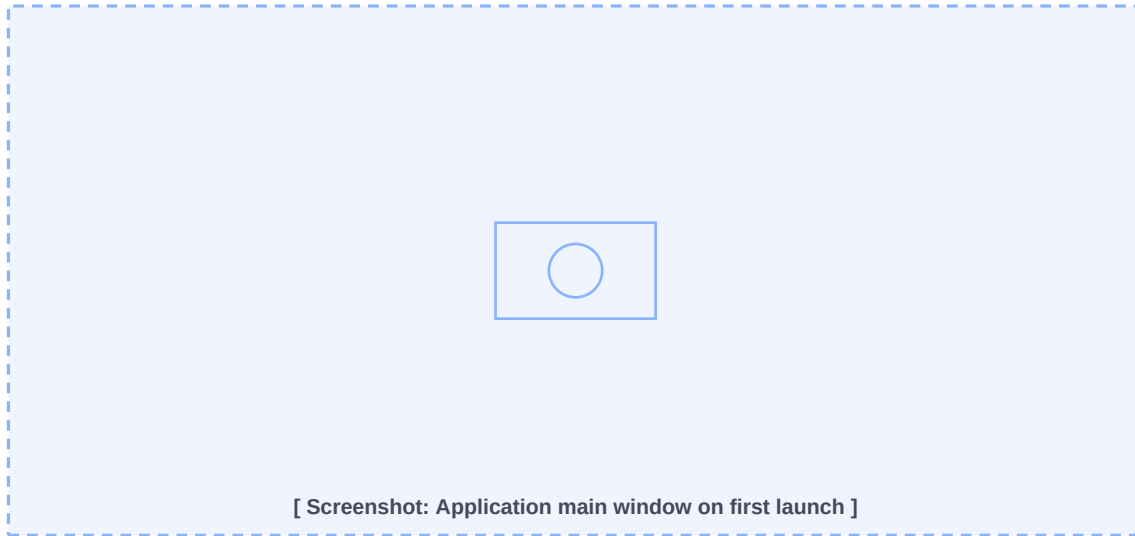


Fig 8.1 - Auto Installer Genius on first launch

Optional: Create a desktop shortcut

Create a file named **AutoInstaller.bat** on your desktop:

```
@echo off
cd /d F:\MySws\auto-installerV1.0
python main.py
```

Optional: Create a Windows shortcut (.lnk)

- Right-click the desktop and select New - Shortcut.
- Target: python F:\MySws\auto-installerV1.0\main.py
- Start in: F:\MySws\auto-installerV1.0
- Name it: Auto Installer Genius

9. First-Time Setup Checklist

Use this checklist to confirm everything is configured correctly before your first installation run:

	Checklist Item
☐	Python 3.10+ installed and accessible via 'python --version'
☐	pip install -r requirements.txt completed without errors
☐	CEREBRAS_API_KEY set in the .env file
☐	SSH server running on the target Ubuntu machine
☐	SSH user has sudo privileges on the target machine
☐	Network connectivity confirmed (ping or SSH test)

[]	Application launches with 'python main.py'
[]	Entered host details and clicked Install for a test run
[]	Log file created in the logs/ directory after first run

10. Upgrading to a New Version

To upgrade to a newer version of Auto Installer Genius:

If installed via Git:

```
cd F:\MySws\auto-installer
git pull origin main
```

If installed via ZIP download:

- Download the new ZIP from <https://github.com/vishnu-18/auto-installer>
- Extract to a new folder (e.g. auto-installerV1.1).
- Copy your existing .env file to the new folder.
- Copy your .settings.json file to preserve log path and local package dir settings.
- Copy any saved .mc.json configuration files you want to keep.
- Run pip install -r requirements.txt in the new folder.

Note: Your .env file, .settings.json, and .mc.json config files are not overwritten by upgrades. Always back them up before upgrading.

11. Uninstalling the Application

Auto Installer Genius does not use a Windows installer. To remove it, simply delete the application folder:

```
rmdir /s /q F:\MySws\auto-installerV1.0
```

To also remove the Python packages installed for this app (only if not used by other projects):

```
pip uninstall paramiko python-dotenv requests reportlab -y
```

Note: Deleting the folder removes all log files, saved configs, and settings. Back up anything you want to keep before deleting.

12. Troubleshooting

If you encounter issues during setup or first run, refer to the table below:

Error / Symptom	Likely Cause	Fix
ModuleNotFoundError: paramiko	Dependencies not installed	Run: pip install -r requirements.txt
ModuleNotFoundError: tkinter	Python installed without tcl/tk	Reinstall Python, tick 'tcl/tk and IDLE' option
401 Unauthorized (Cerebras API)	Invalid or missing API key	Check CEREBRAS_API_KEY in .env via Settings menu

403 Forbidden (Cerebras API)	API key expired or rate limited	Regenerate key at cloud.cerebras.ai
404 Not Found (Cerebras API)	Wrong model name in .env	Set CEREBRAS_MODEL=qwen-3-235b-a22b-instruct-2507
Error reading SSH protocol banner	Wrong host/port or SSH not running	Verify IP address and run: sudo systemctl status ssh
Authentication failed	Wrong username or password	Double-check credentials in the input fields
Connection timed out	Firewall blocking port 22	Allow port 22 in firewall: sudo ufw allow 22
bash: syntax error near 'do'	Old ssh_client.py version	Ensure ssh_client.py is the latest version from GitHub
sudo: no tty present	SUDO_ASKPASS not set up	Auto-handled on connect - ensure SSH connects successfully first
python not recognized	Python not in Windows PATH	Reinstall Python and check 'Add to PATH' during setup

Checking the log file for details

Every run produces a log file in the logs/ directory. The log contains the full command history, outputs, and AI verdicts. Check it first when diagnosing any failure:

```
F:\MySws\auto-installerV1.0\logs\install_SoftwareName_YYYYMMDD_HHMMSS.log
```

Getting further help

- Check the GitHub repository for known issues and updates:
- <https://github.com/vishnu-18/auto-installer>
- Review the User Manual for detailed usage instructions.